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1 Decay Scheme

Mn-52 decays via electron capture and beta plus to various excited levels in Cr-52.
Le Mn-52 se désintègre par capture électronique et par bêta plus vers plusieurs niveaux excités du Cr-52.

2 Nuclear Data

$T_{1/2}({}^{52}\text{Mn})$: 5,592 (3) d
 $Q^+({}^{52}\text{Mn})$: 4712,0 (19) keV

2.1 Electron Capture Transitions

	Energy (keV)	Probability (%)	Nature	lg <i>ft</i>	<i>P_K</i>	<i>P_L</i>	<i>P_M</i>
ε _{0,8}	84,9 (19)	0,0027 (6)	2nd Forbidden	5,73	0,797 (6)	0,175 (5)	0,0262 (8)
ε _{0,7}	696,5 (19)	1,07 (5)	Allowed	6,587	0,88463 (34)	0,09883 (23)	0,01538 (9)
ε _{0,6}	1096,1 (19)	7,67 (6)	Allowed	6,13	0,88519 (34)	0,09835 (23)	0,01529 (9)
ε _{0,5}	1296,7 (19)	0,025 (38)	2nd Forbidden	9,6	0,88429 (35)	0,09913 (24)	0,01542 (9)
ε _{0,4}	1598,1 (19)	59,9 (7)	Allowed	5,6	0,88550 (34)	0,09810 (23)	0,01525 (9)
ε _{0,3}	1944,2 (19)	0,02 (7)	2nd Forbidden	10,4	0,88496 (34)	0,09855 (23)	0,01532 (9)
ε _{0,2}	2342,3 (19)	0,2 (6)	2nd Forbidden	9,8	0,88518 (34)	0,09837 (23)	0,01529 (9)

2.2 β⁺ Transitions

	Energy (keV)	Probability (%)	Nature	lg <i>ft</i>
β _{0,6} ⁺	74,1 (19)	0,00133 (17)	Allowed	6,13
β _{0,5} ⁺	274,7 (19)	0,00009 (13)	2nd Forbidden	9,6
β _{0,4} ⁺	576,1 (19)	30,85 (43)	Allowed	5,6
β _{0,3} ⁺	922,2 (19)	0,011 (26)	2nd Forbidden	10,4
β _{0,2} ⁺	1320,3 (19)	0,3 (6)	2nd Forbidden	9,8

2.3 Gamma Transitions and Internal Conversion Coefficients

	Energy (keV)	P _{γ+ce} (%)	Multipolarity	α _K (10 ⁻³)	α _L (10 ⁻⁴)	α _M (10 ⁻⁵)	α _T (10 ⁻³)	α _π (10 ⁻⁵)
γ _{6,5} (Cr)	200,580 (38)	0,0772 (22)	M1+E2	15 (11)	14 (11)	18 (14)	16 (12)	
γ _{4,3} (Cr)	346,030 (33)	0,988 (31)	E2	3,43 (5)	3,23 (5)	4,24 (6)	3,79 (6)	
γ _{3,2} (Cr)	398,090 (31)	0,089 (6)	M1+E2	1,4 (7)	1,4 (7)	1,8 (9)	1,6 (8)	
γ _{7,6} (Cr)	399,542 (47)	0,182 (6)	M1+E2	1,4 (7)	1,3 (7)	1,8 (9)	1,6 (8)	
γ _{6,4} (Cr)	501,970 (34)	0,185 (25)	M1+E2	0,7 (3)	0,67 (25)	0,9 (4)	0,8 (3)	
γ _{7,5} (Cr)	600,42 (5)	0,414 (29)	M1+E2	0,44 (13)	0,41 (12)	0,54 (16)	0,48 (14)	
γ _{5,3} (Cr)	647,541 (38)	0,410 (14)	M1+E2	0,27 (18)	0,25 (17)	0,32 (23)	0,29 (20)	
γ _{4,2} (Cr)	744,219 (32)	90,0 (8)	E2	0,302 (5)	0,281 (4)	0,370 (6)	0,334 (5)	
γ _{6,3} (Cr)	848,141 (33)	3,361 (3)	M1+E2	0,18 (3)	0,17 (3)	0,22 (4)	0,20 (4)	
γ _{7,4} (Cr)	901,880 (47)	0,044 (4)	M1+E2	0,158 (24)	0,147 (23)	0,19 (3)	0,17 (3)	
γ _{2,1} (Cr)	935,528 (27)	94,904 (30)	E2	0,1658 (24)	0,1536 (22)	0,202 (3)	0,183 (3)	
γ _{5,2} (Cr)	1045,730 (37)	0,07 (2)	M1+E2	0,113 (13)	0,105 (13)	0,138 (16)	0,125 (15)	
γ _{6,2} (Cr)	1246,380 (32)	4,229 (40)	M1+E2	0,078 (7)	0,072 (7)	0,095 (8)	0,101 (10)	1,5 (3)
γ _{7,3} (Cr)	1247,830 (46)	0,38 (4)	M1+E2	0,078 (7)	0,072 (6)	0,095 (8)	0,101 (10)	1,5 (3)
γ _{3,1} (Cr)	1333,632 (29)	5,08 (3)	E2	0,0731 (11)	0,0674 (10)	0,0887 (13)	0,1175 (17)	3,68 (6)
γ _{1,0} (Cr)	1434,080 (17)	100,0000 (19)	E2	0,0627 (9)	0,0578 (8)	0,0760 (11)	0,1340 (19)	6,47 (9)
γ _{7,2} (Cr)	1645,809 (45)	0,050 (3)	M1+E2	0,0453 (23)	0,0417 (22)	0,055 (3)	0,181 (20)	13,1 (18)
γ _{5,1} (Cr)	1981,110 (34)	0,035 (3)	E2	0,0334 (5)	0,0307 (5)	0,0404 (6)	0,341 (5)	30,4 (5)
γ _{8,2} (Cr)	2257,41 (20)	0,0027 (6)	M1+E2	0,0257 (8)	0,0236 (8)	0,0311 (10)	0,43 (4)	40 (4)

3 Atomic Data

3.1 Cr

ω _K	:	0,289	(5)
ω _L	:	0,0045	(9)
n _{KL}	:	1,508	(5)

3.1.1 X Radiations

	Energy (keV)	Relative probability
X _K		
Kα ₂	5,40557	50,91
Kα ₁	5,41479	100
Kβ ₁	5,94677	} 20,31249
Kβ ₅ ''	5,987	
X _L		
Lℓ	0,5003	
Lα	0,5729 - 0,57695	
Lη	0,5102	
Lβ	0,57515 - 0,69748	
Lγ	0,58496 - 0,58496	

3.1.2 Auger Electrons

	Energy (keV)	Relative probability
Auger K		
KLL	4,554 - 4,794	100
KLX	5,206 - 5,412	26,9
KXY	5,841 - 5,985	1,81
Auger L 0,4263 - 0,6920		

4 Electron Emissions

		Energy (keV)	Electrons (per 100 disint.)
e _{AL}	(Cr)	0,4263 - 0,6920	6,74 (10)
e _{AK}	(Cr)		
	KLL	4,554 - 4,794	} 43,4 (7)
	KLX	5,206 - 5,412	
	KXY	5,841 - 5,985	
ec _{6,5} T	(Cr)	194,59 - 200,58	0,0012 (9)
ec _{4,3} T	(Cr)	340,04 - 346,03	0,00373 (13)
ec _{3,2} T	(Cr)	392,10 - 398,09	0,00014 (7)
ec _{7,6} T	(Cr)	393,553 - 399,542	0,00029 (15)
ec _{6,4} T	(Cr)	495,98 - 501,97	0,00015 (6)
ec _{7,5} T	(Cr)	594,43 - 600,42	0,00020 (6)
ec _{5,3} T	(Cr)	641,552 - 647,541	0,00012 (8)
ec _{4,2} T	(Cr)	738,230 - 744,219	0,0301 (5)
ec _{6,3} T	(Cr)	842,152 - 848,141	0,00067 (13)
ec _{2,1} T	(Cr)	929,539 - 935,528	0,01736 (28)
ec _{6,2} T	(Cr)	1240,39 - 1246,38	0,000427 (42)
ec _{3,1} T	(Cr)	1327,643 - 1333,632	0,000597 (9)
ec _{1,0} T	(Cr)	1428,09 - 1434,08	0,01340 (19)
$\beta_{0,6}^+$	max:	74,1 (19)	} 0,00133 (17)
	avg:	34,1 (8)	
$\beta_{0,4}^+$	max:	576,1 (19)	} 30,85 (43)
	avg:	241,6 (8)	
$\beta_{0,5}^+$	max:	274,7 (19)	} 0,00009 (13)
	avg:	132,2 (9)	
$\beta_{0,3}^+$	max:	922,2 (19)	} 0,011 (26)
	avg:	418,1 (8)	
$\beta_{0,2}^+$	max:	1320,3 (19)	} 0,3 (6)
	avg:	594,5 (8)	

5 Photon Emissions

5.1 X-Ray Emissions

		Energy (keV)	Photons (per 100 disint.)		
XL	(Cr)	0,5003 - 0,69748	0,030 (7)		
XK α_2	(Cr)	5,40557	5,25 (12)	}	K α
XK α_1	(Cr)	5,41479	10,30 (23)		
XK β_1	(Cr)	5,94677	2,09 (5)	}	K' β_1
XK β_5''	(Cr)	5,987			K' β_2

5.2 Gamma Emissions

	Energy (keV)	Photons (per 100 disint.)
$\gamma_{6,5}(\text{Cr})$	200,58 (4)	0,076 (2)
$\gamma_{4,3}(\text{Cr})$	346,03 (3)	0,984 (31)
$\gamma_{3,2}(\text{Cr})$	398,09 (9)	0,089 (6)
$\gamma_{7,6}(\text{Cr})$	399,542 (19)	0,182 (6)
$\gamma_{6,4}(\text{Cr})$	501,97 (8)	0,185 (25)
$\gamma^\pm$	511	62,3 (15)
$\gamma_{7,5}(\text{Cr})$	600,42 (23)	0,414 (29)
$\gamma_{5,3}(\text{Cr})$	647,537 (30)	0,410 (14)
$\gamma_{4,2}(\text{Cr})$	744,213 (6)	90,0 (8)
$\gamma_{6,3}(\text{Cr})$	848,134 (26)	3,36 (3)
$\gamma_{7,4}(\text{Cr})$	901,87 (18)	0,044 (4)
$\gamma_{2,1}(\text{Cr})$	935,519 (10)	94,887 (30)
$\gamma_{5,2}(\text{Cr})$	1045,72 (8)	0,07 (2)
$\gamma_{6,2}(\text{Cr})$	1246,36 (12)	4,229 (40)
$\gamma_{7,3}(\text{Cr})$	1247,81 (12)	0,38 (4)
$\gamma_{3,1}(\text{Cr})$	1333,614 (11)	5,079 (30)
$\gamma_{1,0}(\text{Cr})$	1434,060 (15)	99,9866 (19)
$\gamma_{7,2}(\text{Cr})$	1645,781 (29)	0,050 (3)
$\gamma_{5,1}(\text{Cr})$	1981,07 (3)	0,035 (3)
$\gamma_{8,2}(\text{Cr})$	2257,36 (19)	0,0027 (6)

6 Main Production Modes

Cr – 52(d,2n)Mn – 52 σ : 0,15 barns

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